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$$\begin{aligned} & \int \frac{dx}{2x^2 + 2x + 1} \\ &= \frac{1}{2} \int \frac{dx}{x^2 + x + \frac{1}{2}} \\ &= \frac{1}{2} \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{1}{4}} \\ & \text{Stel } t = x + \frac{1}{2} \text{ dan } dt = dx \\ &= \frac{1}{2} \int \frac{dt}{t^2 + \left(\frac{1}{2}\right)^2} \\ &= \frac{1}{2} \cdot 2 \operatorname{Bgtan} \left(\frac{x + \frac{1}{2}}{\frac{1}{2}} \right) + c \\ &= \operatorname{Bgtan}(2x + 1) + c \end{aligned}$$

References